

### Tutorial 3

1 a)  $y'' - 3y' + 2y = 0$

Set  $y = e^{\lambda x}$

Then  $\lambda^2 e^{\lambda x} - 3\lambda e^{\lambda x} + 2e^{\lambda x} = 0$

so  $(\lambda^2 - 3\lambda + 2)e^{\lambda x} = 0$

so  $\lambda^2 - 3\lambda + 2 = 0$  so  $(\lambda - 1)(\lambda - 2) = 0$

$\lambda = 1$  or  $\lambda = 2$

General solution:  $y = C_1 e^x + C_2 e^{2x}$

b)  $2y'' + 8y' + 6y = 0$

$y'' + 4y' + 3y = 0$

Set  $y = e^{\lambda x}$

$(\lambda^2 + 4\lambda + 3)(e^{\lambda x}) = 0$

$(\lambda + 3)(\lambda + 1)(e^{\lambda x}) = 0$

so  $\lambda = -3$  or  $\lambda = -1$

General solution:  $y = C_1 e^{-3x} + C_2 e^{-x}$

c)  $y'' - 2y' + y = 0$  Try  $y = e^{\lambda x}$

Then  $(\lambda - 1)^2 e^{\lambda x} = 0$

so  $\lambda = 1$ . One real root Try  $y = x e^{\lambda x}$

general solution:

$y = C_1 e^x + C_2 x e^x$

$$d) \quad y'' + 3y' + 5y = 0$$

$$\text{Set } y = e^{\lambda x}$$

$$\text{Then } (\lambda^2 + 3\lambda + 5)(e^{\lambda x}) = 0$$

$$\text{so } \lambda^2 + 3\lambda + 5 = 0$$

$$\text{so } \lambda = \frac{-3 \pm \sqrt{9 - 4(5)}}{2} = \frac{-3 \pm \sqrt{-11}}{2}$$

$$= \frac{-3 \pm i\sqrt{11}}{2} = -\frac{3}{2} \pm i\omega$$

$$\text{Here } a=3, b=5 \quad (y'' + ay' + by = 0)$$

$$\text{and } \omega^2 = b - \frac{a^2}{4} = 5 - \frac{9}{4} = \frac{11}{4}$$

$$\text{so } \omega = \frac{\sqrt{11}}{2}$$

$$y_1 = e^{-\frac{3x}{2}} \left[ A \cos\left(\frac{\sqrt{11}x}{2}\right) + B \sin\left(\frac{\sqrt{11}x}{2}\right) \right]$$

### Tutorial 3 ctd.

$$2.a) y'' - 7y' + 10y = 0 \quad y(0) = 1, y'(0) = 0$$

$$\text{Set } y = e^{\lambda x}$$

$$(\lambda^2 - 7\lambda + 10)e^{\lambda x} = 0$$

$$(\lambda - 5)(\lambda - 2) = 0 \quad \text{so } \lambda = 5 \text{ or } \lambda = 2$$

$$\text{General solution: } y = C_1 e^{5x} + C_2 e^{2x}$$

$$\text{Initial values: } y(0) = 1 = C_1 + C_2$$

$$y'(0) = 5C_1 + 2C_2 = 0$$

$$2C_1 + 2C_2 = 0$$

$$\text{so } C_1 = 0, C_2 = 1 \text{ and } y(x) = e^{2x}$$

$$b) y'' + 6y' + 9 = 0 \quad y(0) = 0, y'(0) = -1$$

$$\text{Set } y = e^{\lambda x}, (\lambda + 3)^2 e^{\lambda x} = 0 \quad \text{so } \lambda = -3$$

$$\text{so general solution is } y(x) = C_1 e^{-3x} + C_2 x e^{-3x}$$

$$y(0) = 0 = C_1$$

$$y'(x) = C_2 e^{-3x} + 3C_2 x e^{-3x}$$

$$y'(0) = C_2 = -1$$

$$\text{so } y(x) = -x e^{-3x}$$

$$2c) \quad y'' + 5y' + 7y = 0$$

$$\text{Set } y = e^{\lambda x} \text{ then } (\lambda^2 + 5\lambda + 7)(e^{\lambda x}) = 0$$

$$\text{so } \lambda^2 + 5\lambda + 7 = 0 \quad \text{so } \lambda = \frac{-5 \pm \sqrt{25 - 4(7)}}{2}$$

$$\lambda = \frac{-5 \pm i\sqrt{3}}{2}$$

2.

$$\text{so } y(x) = e^{-5x/2} \left[ A \cos\left(\frac{\sqrt{3}x}{2}\right) + B \sin\left(\frac{\sqrt{3}x}{2}\right) \right]$$

$$y(0) = 1 \quad \text{so} \quad A = 1$$

$$y'(x) = -\frac{5}{2} e^{-5x/2} \left[ A \cos\left(\frac{\sqrt{3}x}{2}\right) + B \sin\left(\frac{\sqrt{3}x}{2}\right) \right] \\ + e^{-5x/2} \frac{\sqrt{3}}{2} \left[ -A \sin\left(\frac{\sqrt{3}x}{2}\right) + B \cos\left(\frac{\sqrt{3}x}{2}\right) \right]$$

$$y'(0) = 0 = -\frac{5}{2} [A + B \cdot 0] + \frac{\sqrt{3}}{2} [-A \cdot 0 + B]$$

$$0 = -\frac{5}{2} + \frac{B\sqrt{3}}{2}$$

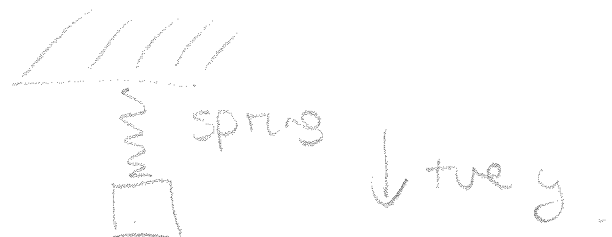
$$\text{so } B\sqrt{3} = 5$$

$$B = 5/\sqrt{3}$$

$$y(x) = e^{-5x/2} \left[ \cos\left(\frac{\sqrt{3}x}{2}\right) + \frac{5}{\sqrt{3}} \sin\left(\frac{\sqrt{3}x}{2}\right) \right]$$

# Tutorial 3 chd

3.  $F_s = -ky$  (spring force)



$F_d = -cy'$  (damper force)

$F_s$ : restoring force,  $k$  is spring constant.

Damper: (or dashpot) slows the motion of the spring

(eg to slow the closing of a door).

If  $y' > 0$ ,  $F_d$  acts to slow the mass  
ie acceleration is negative.

If  $y' < 0$ ,  $F_d$  gives positive acceleration.

$$F_s + F_d = F = ma = my''$$

$$\text{so } my'' = -ky - cy'$$

$$my'' + cy' + ky = 0$$

Here  $m, k > 0$  and  $c \geq 0$ .

Say  $c > 0$ , then

$$y'' + \frac{c}{m}y' + \frac{k}{m}y = 0$$

Set  $y = e^{\lambda x}$ , then  $\lambda^2 + \frac{c}{m}\lambda + \frac{k}{m} = 0$

$$\text{so } \lambda = \frac{-\frac{c}{m} \pm \sqrt{\left(\frac{c}{m}\right)^2 - 4\frac{k}{m}}}{2} = \frac{-\frac{c}{m} \pm \sqrt{\frac{c^2 - 4km}{m^2}}}{2}$$

$$\lambda = \frac{-\frac{c}{m} \pm \sqrt{\frac{c^2 - 4km}{m^2}}}{2} = -\alpha \pm \beta$$

$$\alpha = \frac{c}{2m} \quad \text{and} \quad \beta = \frac{\sqrt{c^2 - 4mk}}{2m}$$

Case 1 Overdamping,  $c^2 - 4mk > 0$

$$y = c_1 e^{-(\alpha+\beta)t} + c_2 e^{-(\alpha-\beta)t}$$

Since  $c^2 > 4mk$ , then  $\alpha > \beta > 0$

and so  $\alpha + \beta > 0$  and  $\alpha - \beta > 0$

so as  $t \rightarrow \infty$  our solution  $y(t) \rightarrow 0$   
due to the negative exponents.

Case 2 Critical damping

$$y = (c_1 + c_2 t) e^{-\alpha t} \quad \text{since } \beta = 0.$$

as  $t \rightarrow \infty$ , since  $\alpha > 0$ ,  $e^{-\alpha t} \rightarrow 0$

and  $te^{-\alpha t} \rightarrow 0$  too so  $y \rightarrow 0$

Case 3 Underdamped.  $c^2 - 4mk < 0$   
 $c^2 < 4mk$

$$\lambda_1 = -\alpha + i\omega \quad \text{and} \quad \lambda_2 = -\alpha - i\omega$$

$$\text{where } \omega = \frac{\sqrt{4mk - c^2}}{2m} > 0 \quad \text{and} \quad \alpha = \frac{c}{2m} > 0$$

$$y = e^{-\frac{cx}{2m}} \left[ A \cos(\omega t) + B \sin(\omega t) \right]$$

$\underbrace{\hspace{100px}}_{\text{decay}}$ 
 $\underbrace{\hspace{100px}}_{\text{oscillatory}}$

$$\lim_{t \rightarrow \infty} y(t) = 0$$

5.  $m = 4$ ,  $k = 7$ , find  $c$ , so critically damped.

$$c^2 - 4mk = 0$$

$$c^2 = 4(4)(7)$$

$$\text{so } c = 4\sqrt{7}$$

$$y = (c_1 + c_2 t) e^{-\alpha t} \quad \text{where } \alpha = \frac{c}{2m}$$
$$= \frac{4\sqrt{7}}{8} = \frac{\sqrt{7}}{2}$$

$$y = (c_1 + c_2 t) e^{-\sqrt{7}t/2}$$

e.g. let  $y(0) = 1$ , so  $c_1 = 1$

Then need  $c_2 < 0$  to obtain a crossing:

$$1 + c_2 t = 0 \quad c_2(t) = -1$$
$$t = -\frac{1}{c_2} > 0$$

$$y\left(-\frac{1}{c_2}\right) = 0$$

Velocity at  $t = 0$ ?

$$y' = c_2 e^{-\sqrt{7}t/2} + (c_1 + c_2 t) \left(-\frac{\sqrt{7}}{2}\right) e^{-\sqrt{7}t/2}$$

$$y'(0) = c_2 - \frac{\sqrt{7}c_1}{2} = c_2 - \frac{\sqrt{7}}{2} < 0$$

$$y'(0) \text{ must be } < -\frac{\sqrt{7}}{2} \quad \text{as } c_2 < 0$$

4.  $m=1$ ,  $c=1$ ,  $k=0$   
mass, damper system

$$my'' + cy' = 0$$

$$y'' + \frac{c}{m} y' = 0$$

Say  $y = e^{\lambda x}$  then  $\lambda^2 + \frac{\lambda c}{m} = 0$

so  $\lambda = 0$  or  $\lambda = -\frac{c}{m}$

$$y = C_1 e^{-\frac{c}{m}x} + C_2 e^0 = C_1 e^{-x} + C_2$$

$$y(0) = C_1 + C_2 = 1$$

$$y'(0) = -C_1 e^0 = -C_1 = \alpha$$

so  $C_1 = -\alpha$

$$C_2 = 1 - C_1 = 1 + \alpha$$

so  $y = -\alpha e^{-x} + 1 + \alpha$

if  $\alpha = 0$   $y(t) = 1$ , no motion.

if  $\alpha < 0$  say  $\alpha = -1$

$$y(t) = e^{-x}, \text{ decaying to } y(0) < 1.$$

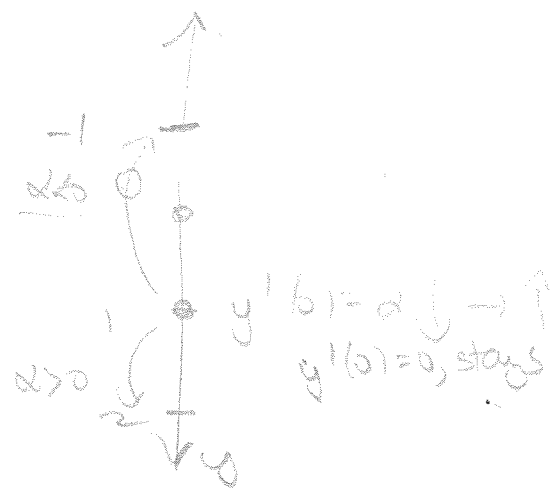
say  $\alpha = -2$   $y(t) = 2e^{-x} - 1$

have  $e^{-x}$  with positive coefficient + some constant, decay.

if  $\alpha > 0$ , say  $\alpha = 1$

$$y(t) = -e^{-x} + 2$$

decay to even greater positive displacement  
then  $y(0) = 1$ .





$$6. \quad \theta'' + \frac{g}{l} \theta = 0.$$

since  $\theta \approx \sin \theta$   
for small  $\theta$ .

$$\text{Set } \theta(t) = e^{\lambda t}$$

$$\text{Then } \left( \lambda^2 + \frac{g}{l} \right) e^{\lambda t} = 0$$

$$\text{so } \lambda^2 = -\frac{g}{l} \quad \text{or } \lambda = \pm i \sqrt{\frac{g}{l}}$$

$$= \pm i\omega \quad \text{where } \omega = \sqrt{\frac{g}{l}}$$

(like mass, spring,  
no damper)

so

$$y = A \cos \sqrt{\frac{g}{l}} t + B \sin \sqrt{\frac{g}{l}} t.$$

oscillatory.

